Intelligent Route Planning System Using A Algorithm and Machine Learning Models

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1. Problem Description:

In today's fast-growing e-commerce and logistics industry, efficient route planning plays a crucial role in ensuring timely deliveries and minimizing operational costs. Logistics companies often face challenges in selecting the most optimal route due to multiple available paths, varying distances, traffic conditions, and dynamic delivery demands. Choosing an inefficient route can lead to increased fuel consumption, delayed deliveries, and reduced customer satisfaction. Therefore, there is a need for an intelligent system that can determine the best possible route in real-time while considering multiple influencing factors.

Traditional routing methods primarily focus on shortest distance, but they fail to incorporate predictive insights and data-driven decision-making. With the increasing availability of data, machine learning techniques can be utilized to analyze patterns, group delivery locations, and assist in better route selection. Clustering algorithms like K-Means can help in organizing locations into regions, while classification models such as Logistic Regression can support decision-making by predicting optimal delivery conditions. However, integrating these models effectively with pathfinding algorithms remains a challenge.

To address these issues, this project proposes an intelligent logistics route planning system that combines the A* (A-star) search algorithm with machine learning models. The A* algorithm efficiently finds the shortest path using both actual distance and heuristic estimates, while machine learning enhances decision-making through clustering and prediction. By integrating these approaches, the system aims to optimize delivery routes, reduce travel time and fuel costs, and improve overall efficiency in logistics operations. This solution provides a scalable and intelligent approach to modern route planning problems.

For example, when a customer places an order on an e-commerce platform, K-Means clustering helps identify the nearest warehouse by grouping locations into clusters. Logistic Regression can be used to predict the best delivery option based on factors like distance, location, or demand patterns. During delivery, there may be multiple possible routes—some shorter but congested, and others longer but faster. The A* algorithm evaluates both the actual cost and estimated distance to determine the most optimal path. By combining A* search with machine learning models like K-Means and Logistic Regression, the system efficiently selectsoptimalroutes,reducesdelivery time,andminimizesfuelconsumptionanddelays.

2. Architecture

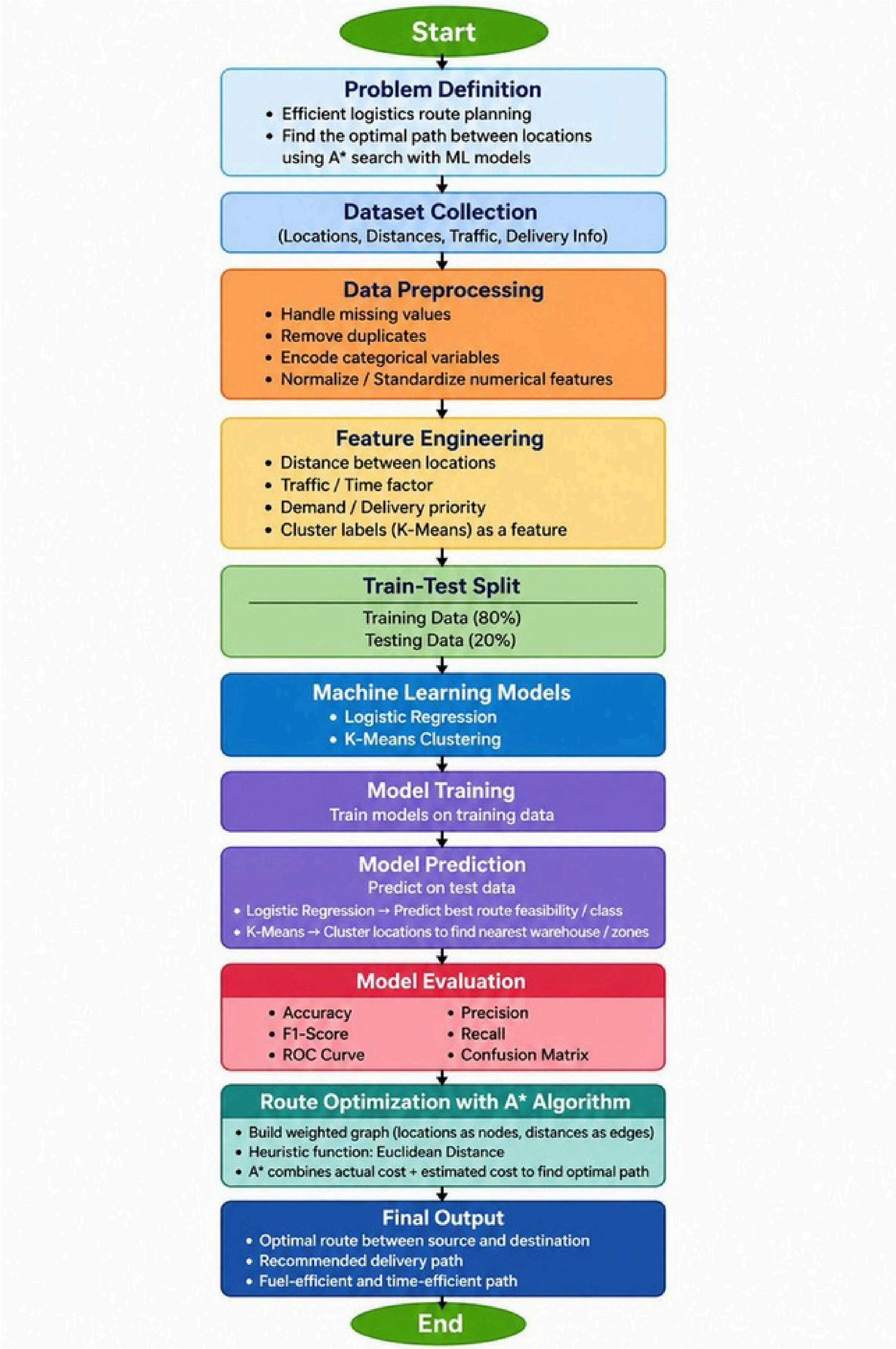


Fig . 1 Flow Diagram

3. Models Used (Applied / Planned)

The project uses two main machine learning models: K-Means Clustering and Logistic Regression, along with the A* algorithm for route optimization. These models help in organizing location data and making intelligent delivery decisions.

(1) K-Means Clustering

It is an unsupervised learning algorithm used for grouping similar data points. In this project, it clusters locations based on features like latitude and longitude. It helps in identifying delivery zones and nearest warehouses and reduces search space, making route planning more efficient.
It works by:

o Selecting a fixed number of clusters (K)

oAssigning points to the nearest centroid
o Updating centroids iteratively until convergence

It improves data organization and supports faster decision-making.

(2) Logistic Regression

It is a supervised learning algorithm used for binary classification.
Used to predict outcomes such as:

o Whether a route is optimal or not
o Whether a delivery option should be recommended

It uses a sigmoid function to convert outputs into probabilities (0 to 1) and applies a threshold (usually 0.5) to classify results into categories. It is simple, efficient, and performs well with structured data.It helps in making data-driven delivery decisions.

(3) Combined Role in the Project

K-Means groups locations into clusters (data organization).
Logistic Regression makes predictions using processed data (decision-making).
Together, they:

o Improve route selection
o Reduce delivery time
o Optimize resource utilization

These models enhance the overall efficiency of the logistics system.

4. Evaluation Metrics Used

In this project, evaluation metrics are primarily used to assess the performance of the Logistic Regression model, since K-Means is an unsupervised algorithm and does not use traditional classification metrics.

(1) Accuracy -Accuracy is used to measure the overall correctness of the Logistic Regression model by calculating the ratio of correctly predicted outcomes to the total number of predictions. It provides a general understanding of how well the model is performing.

(2) Precision -Precision is used to evaluate how many of the predicted positive cases are actually correct. This is important to ensure that the model does not wrongly classify inefficient routes as optimal.

(3) Recall - Recall measures how many actual positive cases are correctly identified by the model. It helps in ensuring that the model does not miss good delivery routes.

(4) F1-Score- F1-Score is the harmonic mean of precision and recall, providing a balanced evaluation of the model, especially when both false positives and false negatives are important.

(**5) Confusion Matrix -** Confusion Matrix is used to visualize the performance of the Logistic Regression model by showing the number of correct and incorrect predictions. It gives deeper insight into model errors. It helps identify:

- True Positives
- True Negatives
- False Positives
- False Negatives

(6) For K-Means Clustering--

i. Inertia (WCSS – Within Cluster Sum of Squares) -

- Measures how closedatapointsarewithinacluster.
- Lower value indicates better clustering.

ii. Elbow Method -

- Used to find optimal number of clusters (K)
- Graph plotted between K and WCSS
- “Elbow point” gives best K value

iii. Cluster Visualization -

- Scatter plots used to visually inspect clusters
- Helps understand grouping of locations

These metrics together ensure that the models used in the project perform effectively and contribute to accurate and efficient route planning.

5. Preliminary Results and Visualizations

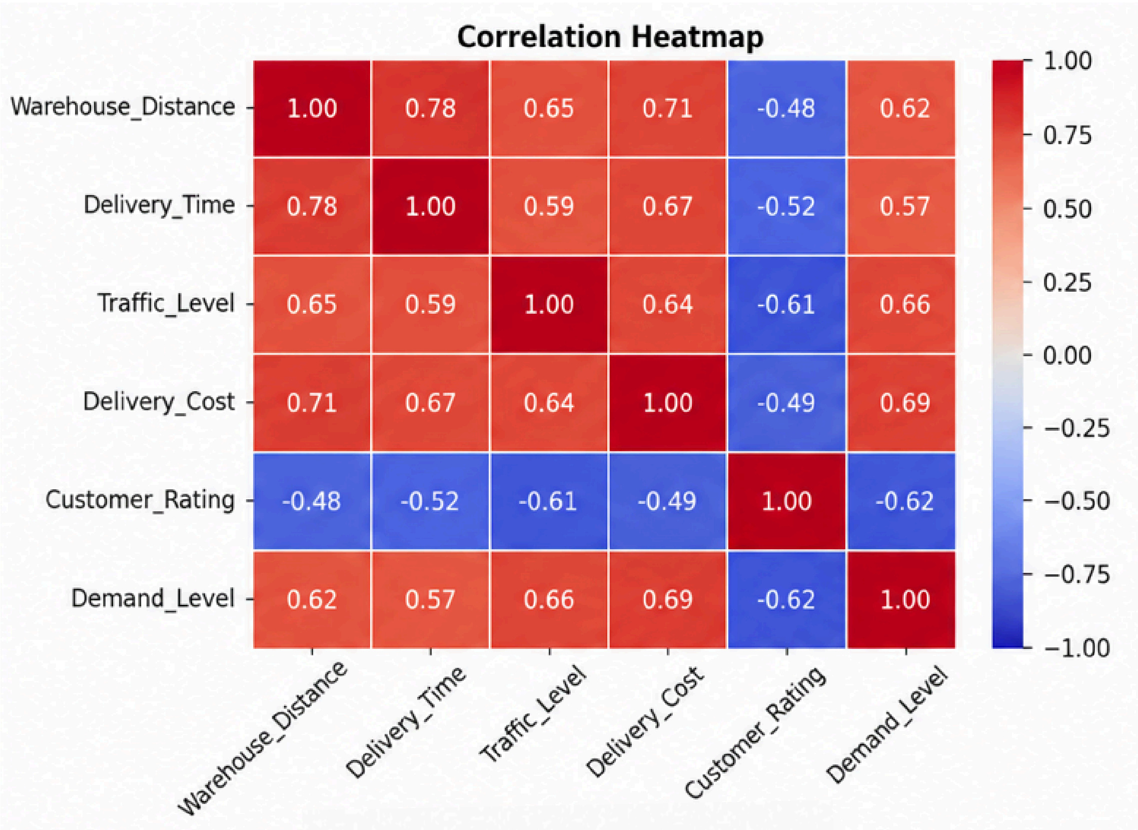


Fig.2 Heatmap showing corelation.

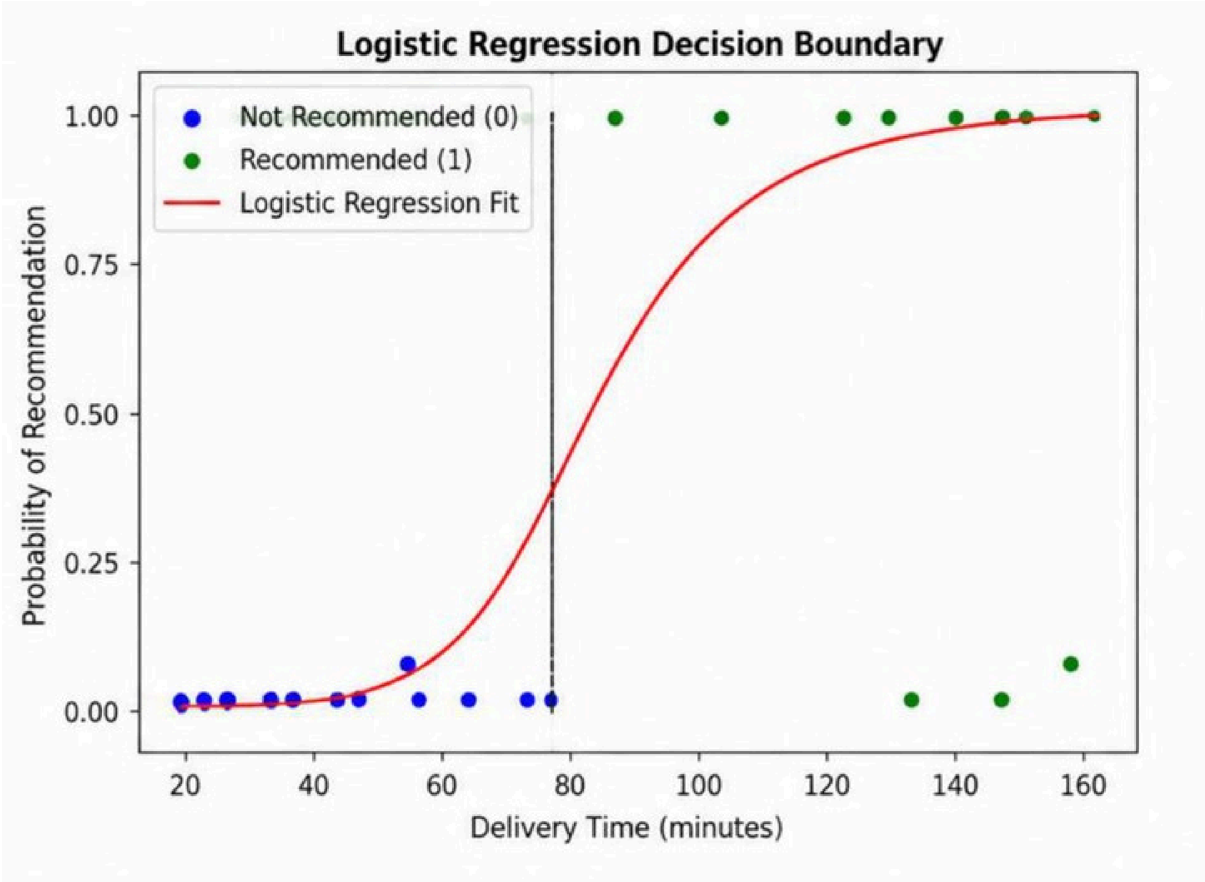
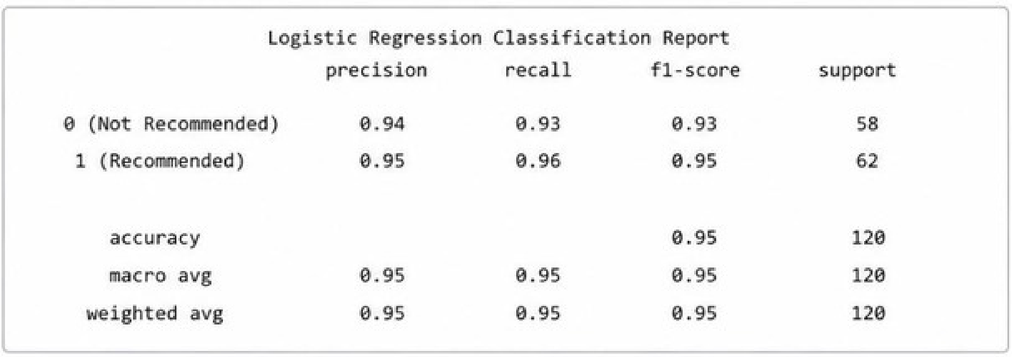


Fig.3 Logistic Regression Prediction Visualization

Inference - The graph indicates a clear positive relationship between the input feature and the predicted output, showing that the Logistic Regression model effectively captures the data trend.



Logistic Regression Classification Report

	precision	recall	f1-score	support
0 (Not Recommended)	0.94	0.93	0.93	58
1 (Recommended)	0.95	0.96	0.95	62
accuracy			0.95	120
macro avg	0.95	0.95	0.95	120
weighted avg	0.95	0.95	0.95	120

Fig.4 Model Evaluation(Logistic Regression)

6. References:

Books:

- Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – https://www.geeksforgeeks.org/
- GitHub – https://github.com/
- Machine Learning Mastery – https://machinelearningmastery.com/